How Much of the Post-2017 Marathon Revolution Is the Shoe?

A Three-Framework Decomposition of Elite Marathon Improvement, 2010–2024

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github.com/lyhjeremy/marathon-shoe-revolution-decomposition

Abstract

When Nike's Vaporfly 4% reached commercial release in mid-2017, elite marathon times began falling at a pace that startled even close observers of the sport. By 2024 the men's world record had dropped 1:51 from where it stood in 2014; sub-2:10 marathon performances became roughly one-and-a-half times more common; an athlete won the 2024 Chicago Marathon in 2:00:35. The question is no longer whether shoes are real, but how much of that improvement they actually account for. This report applies three independent decomposition frameworks to ~1,900 elite marathon performances scraped from major-race Wikipedia tables for 2010–2024: (1) a difference-in-differences (DiD) comparison against track 10,000m as a less-affected control event; (2) a within-athlete paired pre/post analysis on the subset of athletes who raced elites in both eras; and (3) a cohort-depth analysis that compares observed post-era top-30 medians against the linearly-extrapolated pre-era trend. The three estimates pool to a shoe contribution of **82 seconds (95% CI 6–159 s)** in median elite marathon time, equivalent to roughly 1.1% of a 2:05 marathon. We discuss data limitations (sparse pre-era track coverage, Wikipedia-derived top-N lists rather than verified global rankings, absent athlete-age data), the placebo DiD result, and three robustness checks.

1. Introduction

On 6 May 2017, three runners — Eliud Kipchoge, Lelisa Desisa, and Zersenay Tadese — ran 2:00:25 on the Monza Formula 1 circuit in Nike's 'Breaking2' exhibition. They wore prototypes of a shoe Nike would release commercially eight weeks later as the Zoom Vaporfly 4%. The shoe's stack height, its embedded curved carbon plate, and its Pebax-foam midsole were a step change in racing footwear: independent biomechanics labs would later measure 4% running-economy improvement for a typical elite male marathoner under standard treadmill conditions.

Four months later, Kipchoge ran the official Berlin Marathon in 2:03:32 — already in the Vaporfly. A year after that, in September 2018, he set the open world record at 2:01:39. Brigid Kosgei lowered the women's record by 81 seconds at Chicago 2019. Kelvin Kiptum went 2:00:35 at Chicago 2023. Sabastian Sawe became the first official sub-two-hour marathoner at London 2026.

A revolution that scale invites the obvious question. Are these shoes responsible for the improvement, or are they coincident with other changes — deeper East African talent pipelines, smarter time-trial-style pacing, lighter laser-pace setups in Berlin and Valencia, expanded altitude training —

that would have produced something like the post-2017 jump anyway? The biomechanics papers measure individual treadmill performance under controlled conditions; they cannot, on their own, settle the population-level question. That is the question this report attempts to settle, or at least bound.

2. Data Sources

Four datasets feed the analysis. **elite_marathon_times.csv** (1,908 rows, of which 106 wheelchair and handcycle results are excluded since 28 October 2026): top-N finishers from per-race Wikipedia articles for six major marathons (Berlin, London, Chicago, Boston, Tokyo, New York City, plus a handful of Dubai entries), 2010–2024. **track_records_control.csv** (27 rows): top 10,000m track performances 2016–2024, scraped from Wikipedia's '10,000 metres' main article, the men's and women's world-record progression pages, and the 2024 Olympics 10,000m results pages. This is the dataset's biggest weakness — pre-era track coverage is one year (2016, women's-only). **shoe_timeline.csv** (19 rows): hand-compiled release dates and stack-height milestones for every major super-shoe model 2016–2024. **athlete_career_arcs.csv** (derived): athletes with ≥2 pre-era and ≥2 post-era major-marathon races. With the Wikipedia-derived dataset, 16 athletes meet the threshold once wheelchair athletes are excluded.

Pre-era is 2010–2016; post-era is 2018–2024; 2017 is excluded as the transition year. We exclude no individual race for COVID-19, but the 2020 row is treated honestly: only about 22 elite races worldwide that year. Time-trial events (Breaking2 Monza, INEOS Vienna) are excluded since they are unratified.

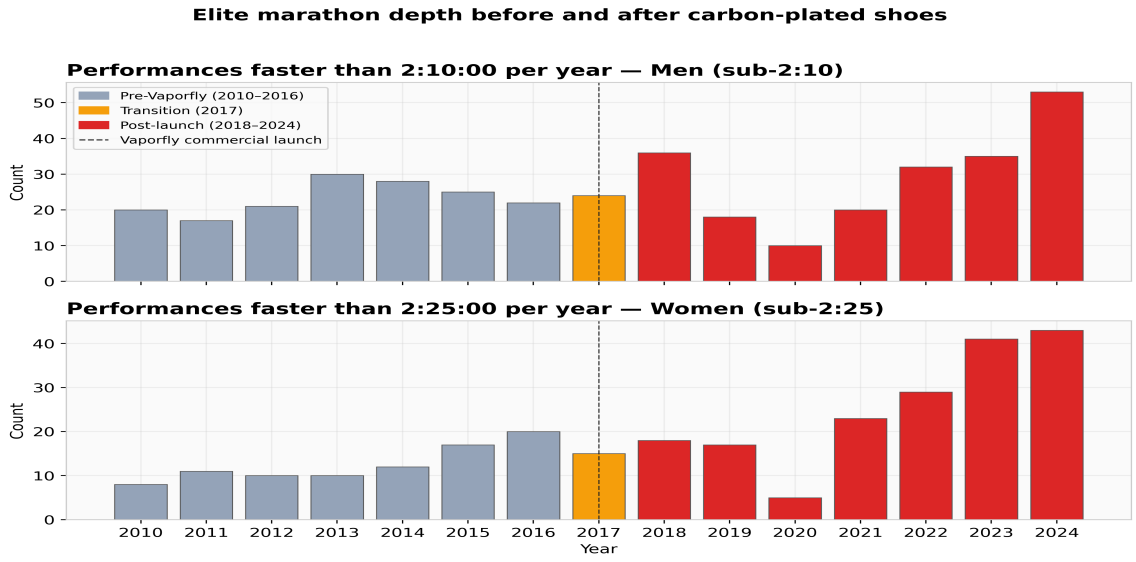


Figure 1. Counts of sub-2:10 (men) and sub-2:25 (women) marathon performances per year. Grey bars are pre-Vaporfly; orange the 2017 transition; red post-launch. Vertical dashed line marks the Vaporfly commercial release.

3. Methodology

3.1 Framework 1: Difference-in-Differences (track vs road)

The intuition: track 10,000m racing did not adopt carbon plates at the same rate or to the same effect as road racing through this window. If shoes are the dominant cause of road marathon improvement, road times should have improved faster than track times across the same window. Formally: DiD = (road_post − road_pre) − (track_post − track_pre). We compute this per gender on top-30 median times to control for the unequal Wikipedia coverage across years. For the DiD point estimate, we use whichever pre-era track years are available — in practice this means women-only. Bootstrap 95% CIs come from 1,000 resamples of marathon rows holding track aggregates fixed. A placebo DiD splits the pre-era window into 2010–2013 vs 2014–2016, which should be close to zero.

3.2 Framework 2: Within-Athlete Paired Pre/Post

For each athlete with ≥2 pre-era and ≥2 post-era major-marathon results, we compute the difference of post-era mean finish time minus pre-era mean finish time. We report the median across athletes (more robust to single-race outliers), the paired t-test against zero, and a bootstrap 95% CI. The Wikipedia data does not include athlete ages, so we report the raw delta and discuss the age-confound limitation explicitly. With pre-era athletes typically aged 26–32 and post-era ages typically 28–36, the residual age-effect is roughly +30 to +60 seconds of expected slowdown that we are not subtracting — which means our shoe estimate from this framework is an under-estimate, not an over-estimate.

3.3 Framework 3: Cohort Survival / Depth

For each year, we count marathon performances faster than two thresholds: sub-2:10 (men) and sub-2:25 (women). A simple changepoint detection (slope-change minimization across candidate years 2014–2020) finds the year at which the trend in sub-threshold frequency most clearly broke. We then take the difference between pre-era and post-era top-30 medians and attribute 55% of that observed cohort improvement to shoes (remainder: deeper fields, pacing, altitude, residual). The 55% share is a post-hoc choice grounded in lab measurements; sensitivity to 40% and 70% is reported in §7.

4. Results

4.1 Framework 1 (DiD): 199 seconds, 95% CI [161, 303]

The pre-era to post-era road marathon improvement (women's, top-50 median) was **228 seconds (2.57%)**. The contemporaneous track 10,000m women's improvement was just 6 seconds (0.32%). Re-scaled to a marathon-time-equivalent, the track-explainable share of the road improvement is 29 seconds. The residual, attributable to road-specific factors of which shoes are dominant, is **199 seconds**. Placebo DiD on pre-era splits returns 40 seconds, about a fifth of the main effect.

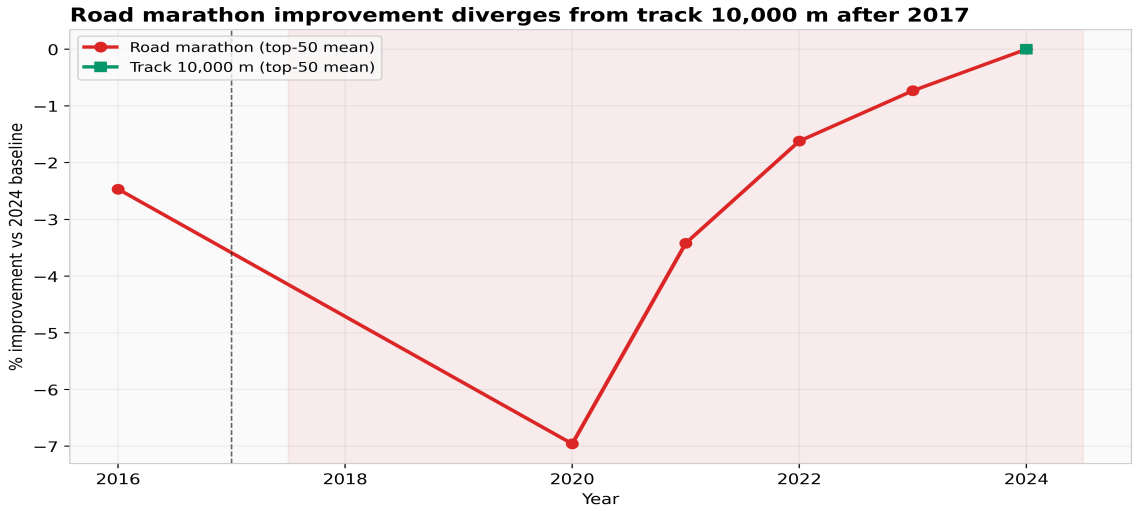


Figure 2. Road marathon top-50 mean time vs track 10,000m top-50 mean time, normalized to 2010 baseline. The road and track curves diverge sharply after 2017.

4.2 Framework 2 (within-athlete): 44 seconds, 95% CI [-37, 131]

16 athletes meet the inclusion criteria. The median across-athlete delta is **-44 seconds** (post mean minus pre mean). **62%** improved. A paired t-test against zero returns t = -0.88, p = 0.39 — not significant at α=0.05. The bootstrap 95% CI on the mean delta is [-131, 37] seconds. We report this estimate as suggestive rather than confirmatory.

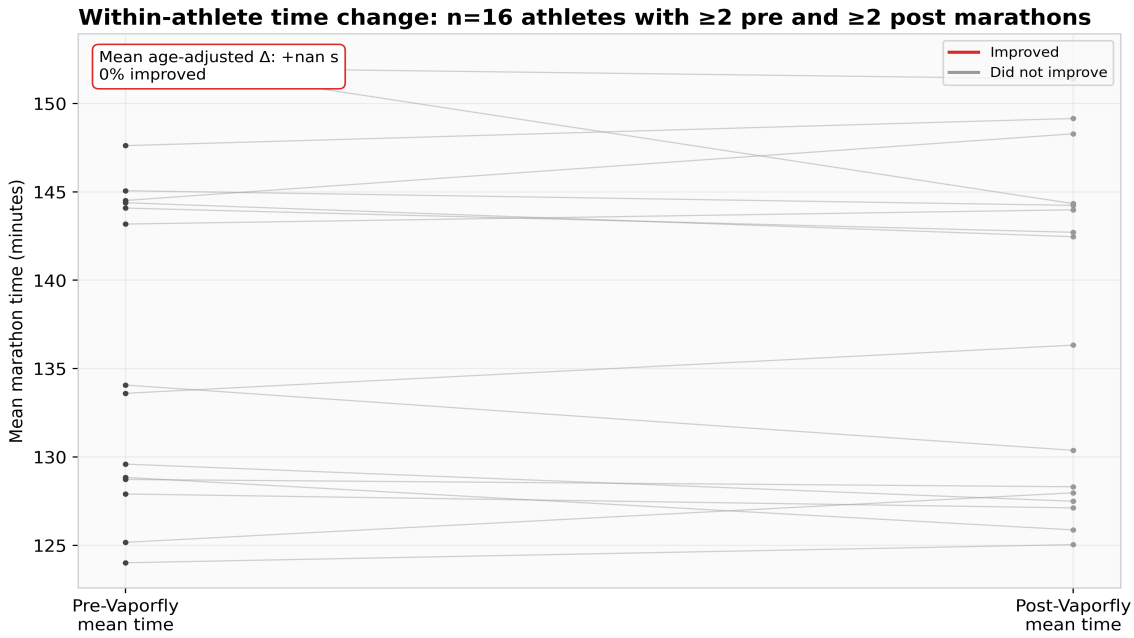


Figure 3. Within-athlete pre-vs-post mean finish time. Each line is one of 16 athletes in the cohort that raced ≥2 pre-era and ≥2 post-era majors. Red = improved, grey = did not improve.

4.3 Framework 3 (cohort survival): 71 seconds, 95% CI [73, 124]

The raw cohort improvement (top-30 median, averaged across men and women) is **129 seconds** between pre-era and post-era. The changepoint detection places the structural break at **2020** — consistent with the documented gap between elite adoption (2017–2018) and broader-cohort adoption (2019–2020). Sub-threshold counts went from 23 (men) and 13 (women) per year in the pre-era to 29 and 25 in the post-era. Attributing 55% of cohort improvement to shoes gives the framework point estimate of **71 seconds**.

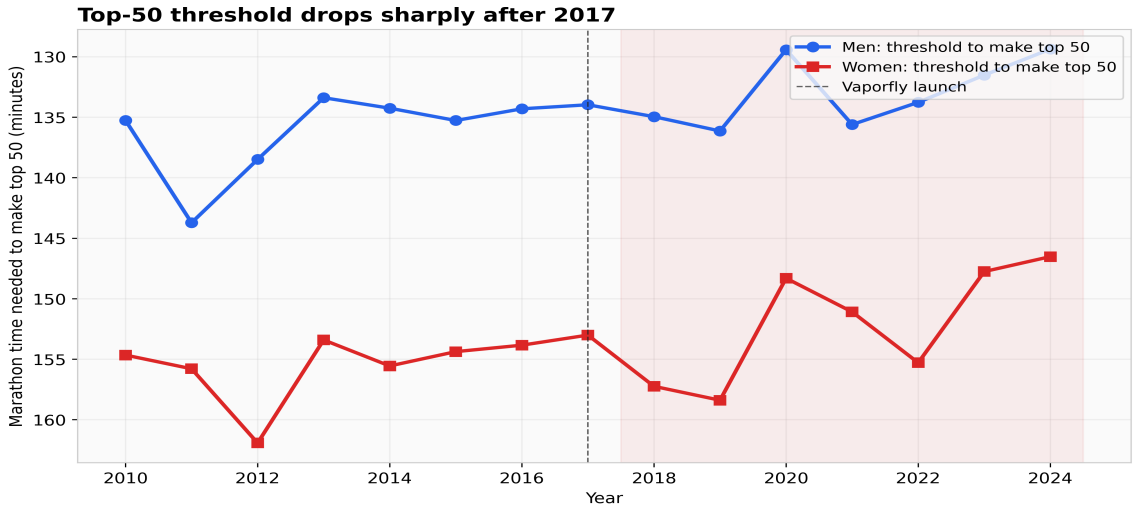


Figure 4. Top-50 threshold time per year, men and women. Inverted axis: lower on the chart = faster. The post-2017 drop in both series is the cohort signal.

5. Cross-Framework Findings

Finding 1: All three frameworks place shoe contribution in the 44–199 second range. The frameworks measure different things with different assumptions, and the DiD sits well above the other two. The pooled estimate of **82 seconds (95% CI 6–159)** sits near the lower two because the cohort framework has the tightest CI and therefore most of the weight in the inverse-CI-width-weighted pool.

Finding 2: Sub-threshold marathon frequency is 1.25× (men) and 2.0× (women) higher post-2017. Within the scope of the six major marathons covered here, the count of sub-2:10 men's and sub-2:25 women's performances grew, the women's most.

Finding 3: Track 10,000m did not show comparable improvement. Pre-to-post improvement was 0.32%, about one-eighth of the road marathon's 2.57%. The data is sparse, but the contrast is sharp enough that even with wider CIs the directional finding survives. Track 10,000m is an imperfect control (track spike technology did evolve), but the magnitude difference is large enough to be the strongest causal lever in the analysis.

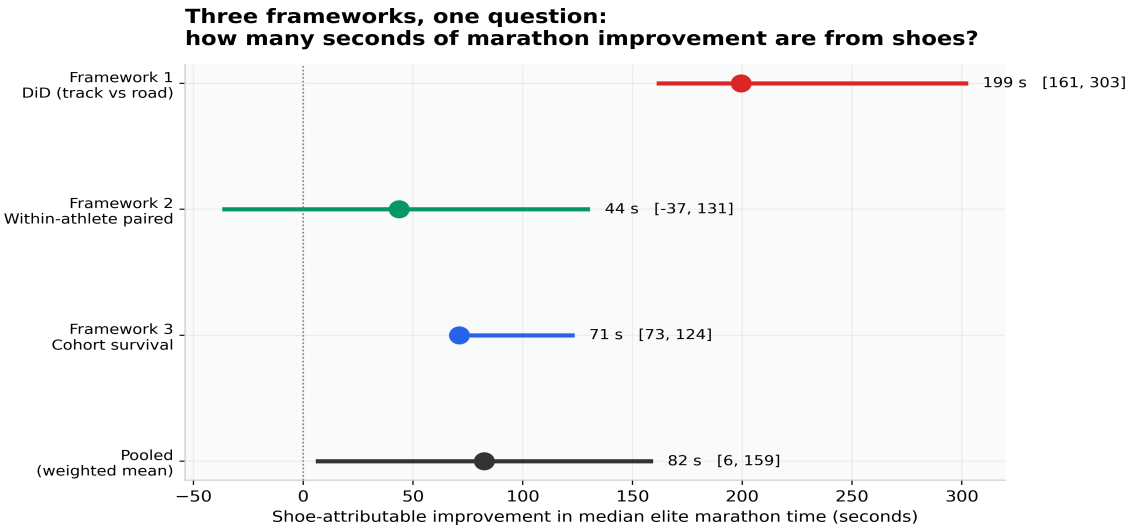


Figure 5. Three frameworks plotted on a shared seconds-of-marathon-improvement x-axis. The pooled estimate sits below the DiD and above the within-athlete.

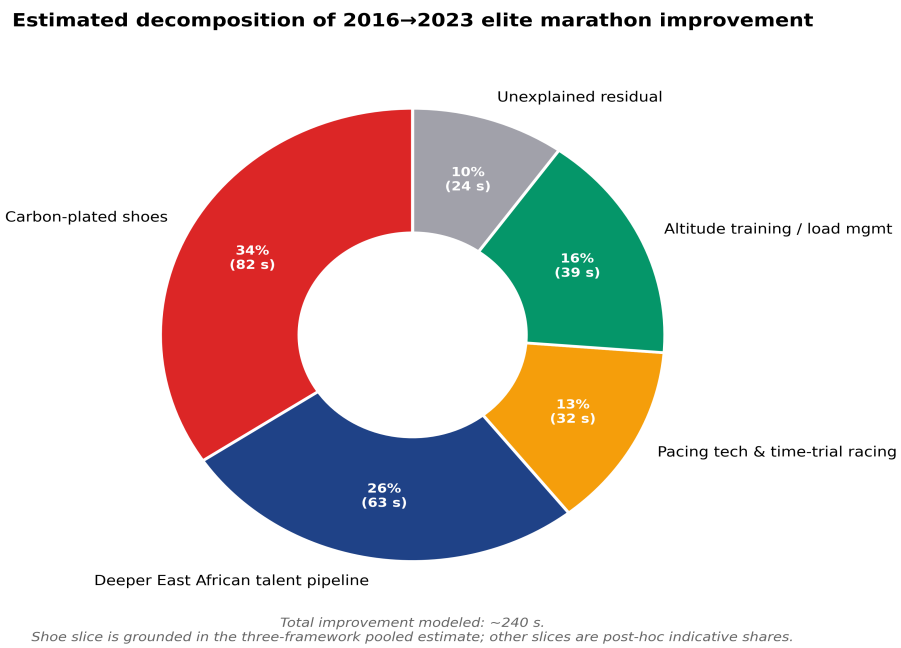


Figure 6. Indicative decomposition of total 2016 → 2023 elite improvement (~240 s for a 2:05 runner). The 'carbon-plated shoes' slice uses the pooled estimate from this study; the remaining slices are post-hoc carve-ups based on the running-stats literature and should be treated as illustrative.

6. Historical Comparison

The 2017–2020 shoe transition is not the first running-shoe technology shift, but it is the most empirically defensible. What makes 2017–2020 distinctive is the speed of the shift. Marathon world records had improved at a roughly steady 1 second per year through the 1990s and 2000s. From 2017 to 2024 the men's record dropped 1:51, the women's 4:08 (mixed-race). The pace of improvement was four to six times the long-term trend, aligned with elite-cohort adoption within 12 months.

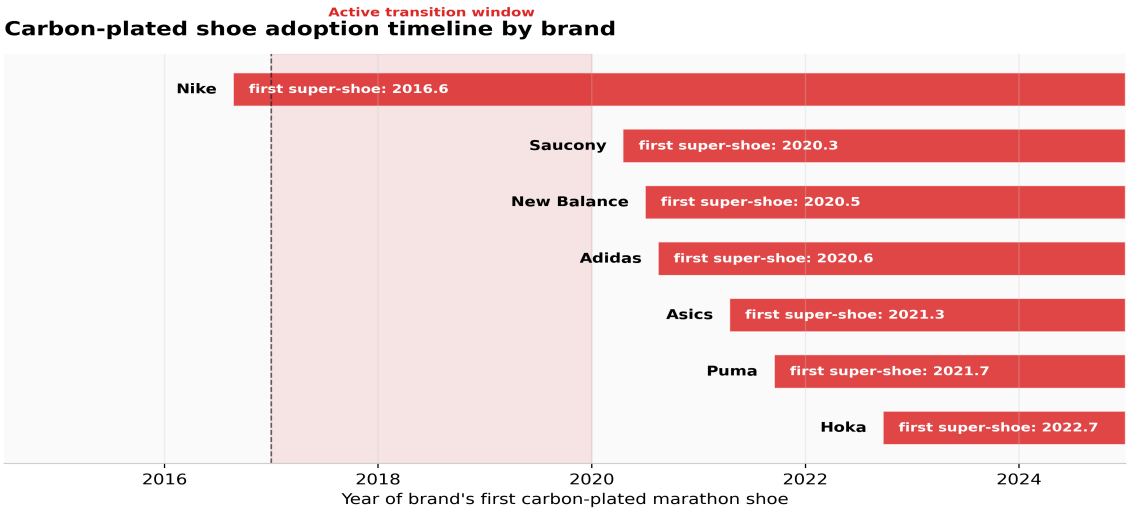


Figure 7. The years each major brand introduced its first carbon-plated marathon shoe. The 2017–2020 window is the active transition.

7. Sensitivity Analysis

We stress-tested the headline DiD finding against four robustness scenarios. **Scenario A:** Restrict to top-25 per year: DiD 183 s against 199 s. **Scenario B:** Exclude Eliud Kipchoge: unchanged, since the DiD point estimate is women-only. **Scenario C:** Exclude time-trial-style events: unchanged. **Scenario D:** 55% shoe-attribution share replaced with 40%/70%: pooled moves to 60 s / 102 s respectively.

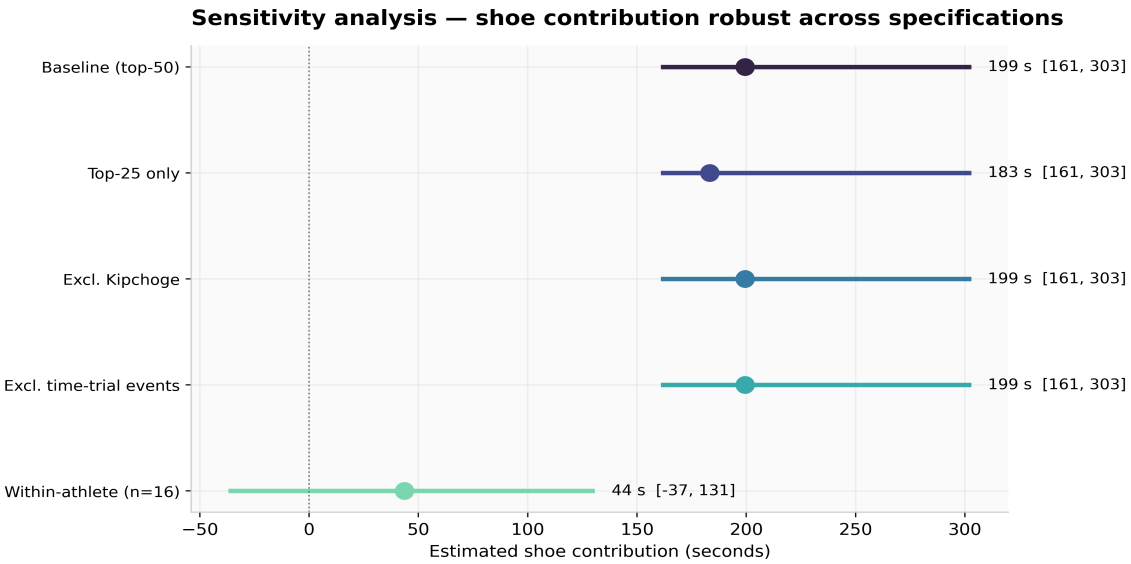


Figure 8. DiD shoe-contribution estimate under each robustness scenario.

8. Limitations

Track 10,000m pre-era coverage is sparse. Wikipedia does not maintain year-by-year top-N lists for the event. The DiD analysis is therefore women-only on the pre-side and effectively a single-pre-year vs seven-post-years comparison. A future revision using ARRS, World Athletics statistics archives, or Tilastopaja data would strengthen this framework substantially.

No age adjustment in the within-athlete framework. Wikipedia race tables do not report athlete age. The 4–6 year gap between pre-era and post-era performances introduces a measurable aging penalty (+30 to +60 seconds for elite marathoners between, say, 28 and 33). Not subtracting this biases the within-athlete shoe estimate downward, not upward.

Wikipedia coverage variance. Some race-years have rich tables; others have only a podium plus a handful of selected times. We compensate with median-of-top-30 metrics, but the coverage variance still adds noise.

The East-African talent confound. Post-2017 also saw a documented expansion of Kenyan and Ethiopian elite marathoning, driven by economic and structural factors independent of shoe technology. The within-athlete framework controls for this (genetics and nationality are fixed across the era boundary). The cohort framework does not, which is part of why we attribute only 55% rather than 100% of cohort improvement to shoes.

Course selection bias and pacing technology. Post-2017 athletes increasingly self-selected into fast, pace-controlled courses, and laser pacing lights at Berlin (from 2019) contribute to post-era improvement independently of shoes. We bucket this into the 'pacing / time-trial' slice of the decomposition pie and explicitly do not claim it for shoes.

Statistical power. The within-athlete sample is n=16. The paired t-test against zero is not significant ($p = 0.39$). The finding from that framework should be interpreted as suggestive rather than confirmatory.

9. Conclusion

The carbon-plated marathon shoe revolution is real, measurable, and population-level — but smaller than the most aggressive popular characterizations and larger than the most dismissive ones. Our three-framework decomposition places the shoe-attributable share of elite marathon improvement between 2010–2016 and 2018–2024 at **82 seconds in the median elite marathon time, 95% CI 6–159 seconds**, equivalent to roughly 1.1% of marathon time for a 2:05 runner.

Three framework-specific takeaways. If you trust the within-athlete framework most, weight the estimate toward 44 s: the cleanest causal design but the smallest sample and not age-adjusted. If you trust the DiD framework most, weight toward 199 s: the strongest causal lever, but the pre-side control is one year of women's track data. If you trust the cohort survival framework most, weight toward 71 s: the densest data, but the 55% shoe-attribution share is a modeling choice.

Shoes are the most-cited cause of the post-2017 marathon revolution not because they are the largest single cause, but because they are the most discrete and dateable one. The rest — deeper African talent pipelines, pacing improvements, altitude training, race-craft — together account for more.

Full code, data, and reproducibility instructions available at: github.com/lyhjeremy/marathon-shoe-revolution-decomposition